

HW5

5.3 b) Find minimum SOP for foic function using a Kmap.

cd \ ce	0	1
00	1	1
01	1	
11		
10	1	

$$F(d, e, f) = \sum m(0, 1, 2, 4)$$

$$F = e'f' + d'f' + d'e'$$

group 1. $\rightarrow 000 + 001 = d'e'(f+f') = d'e'$

group 2. $000 + 100 = e'f'(d+d') = e'f'$

group 3. $000 + 010 = d'f'(e+e') = d'f'$

5.8 b) Find minimum SOP and minimum POS for foic function.

$$F(a, b, c, d) = \sum m(1, 3, 4, 11) + \sum d(2, 7, 8, 12, 14, 15)$$

cd \ ab	00	01	11	10
00	0	1	1	1
01	1	0	0	0
11	1	1	1	1
10	1	0	1	0

group 1. $0100 + 1100 = b'c'd'$ ✓

group 2. $0001 + 0011 = a'b'd$ ✓

group 3. $0011 + 0111 + 1011 + 1111 = b'cd$

this can go away cause the DC's will cancel it out.

Maxterm version

We are doing the problems this way to help with Pattern recognition with DC.

group 1. $0101 + 1101 + 0111 + 1111 = bd = (ac' + cd' + bd + b'd')$

group 2. $0111 + 1111 + 0110 + 1110 = cd = (a+c) \cdot (c+d) \cdot (b'+d') \cdot (b+d)$

group 3. $1100 + 1000 + 1101 + 1001 = ac'$

group 4. $0010 + 1010 + 1000 + 0000 = b'd'$

5.6 c) Find minimum SOP using K-maps + underline essential prime implicants

$$F(a, b, c, d) = \prod M(5, 7, 13, 14, 15)$$

$$\prod D(1, 2, 3, 9)$$

cd \ ab	00	01	11	10
00	1	1	1	1
01	X	5	0	X
11	X	7	0	1
10	X	6	1	0

Note that maxterm 1 is minterm 1!

example:

$$M_0 = A+B+C+D$$

$$m_0 = A'B'C'D'$$

$$m_0' = (A'B'C'D')' = A+B+C+D$$

Maxterms of F correspond to zeros of F

group 1. $1011 + 1010 = b'$

group 2. $0110 + 0100 + 0000 + 0010 = a'd'$

group 3. $0000 + 0100 + 1100 + 1000 = c'd'$ ← these are the only ones that don't change

group 4. $0000 + 1000 = b'c'd'(a+a')$

$$b' + a'd' + c'd' + b'c'd' = 0$$

consensus term / absorption

$$b' + a'd' + c'd' = F(a, b, c, d)$$

← this is cause we are in Maxterm